

HAL-Net: Learning Halation Parameters from Analog Film Highlights

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1 Abstract

Halation, the red-orange glow that forms around bright light sources on analog film, appears in stocks such as CineStill 800T, yet learning-based photo-effect models have largely ignored it. We present HAL-Net, a compact convolutional network that estimates three halation parameters (radius, intensity, warmth) from an image’s highlight regions and pairs them with a physically motivated renderer. HAL-Net extends the parameter-regression approach of FGA-NN (Ameur et al., 2025), which targets film grain for compression codecs, to this separate artifact. No labeled halation dataset exists, so we derive pseudo-labels from 100 curated CineStill 800T scans via a multi-scale highlight-bloom heuristic, then train a 3-layer convolutional network with two residual blocks (693k parameters) to regress those three parameters in under five minutes on a T4 GPU. On a 15-image held-out test split, HAL-Net reaches mean absolute error of 0.015, 0.247, and 0.218 for radius, intensity, and warmth, beating the mean-prediction baseline on the two parameters that drive visual appearance. The model under-predicts intensity on the strongest examples; we trace this to label sparsity at the high end of the intensity range. We release an interactive Gradio demo with manual override sliders so predicted parameters can be checked and adjusted.

2 Introduction

Analog film accumulates a signature set of imperfections during exposure and processing, and two of the most recognizable are grain and halation. Grain is the random luminance and color noise produced by silver halide crystals. FGA-NN introduced the first neural method to analyze it, estimating parameters for grain synthesis pipelines used in video codecs. Halation is a different artifact: the soft, warm-colored glow around bright highlights, caused by light passing through the emulsion, reflecting off the back of the film base, and re-exposing the silver halide layer from behind. CineStill 800T is the best-known carrier of this look. Its anti-halation backing is stripped during processing for cinema use, leaving the glow visible in scans of street lamps, neon signage, and practical lights at night.

Despite the popularity of the CineStill aesthetic, halation has no equivalent of FGA-NN: a tool that takes an arbitrary image and recovers the parameters needed to reproduce the look. Existing halation tools in editing software are fixed presets or raw sliders the user tunes by eye for every image. HAL-Net fills that gap. It maps a photograph directly to three named parameters (radius, intensity, warmth) that a simple renderer turns back into the glow.

HAL-Net is FGA-NN’s idea applied to a different artifact. Where FGA-NN analyzes the statistics of film grain to recover synthesis parameters, HAL-Net analyzes the statistics of highlight regions to recover halation parameters. Both artifacts are components of the same analog look, and a full film-emulation pipeline would want both. Our contributions are:

1. A rule-based pseudo-labeling procedure that turns 100 unlabeled CineStill scans into (radius, intensity, warmth) training targets by measuring color and brightness statistics in the halo region around detected highlights.
2. A lightweight CNN that regresses these three parameters from a 256x256 input image in a single forward pass.
3. A renderer that reconstructs the halation effect from predicted parameters, with manual sliders so any prediction can be overridden.
4. Quantitative evaluation on a 15-image held-out test split, including a mean-prediction baseline and a coverage-cutoff ablation justifying the most consequential heuristic constant.

Because ground truth comes from a heuristic rather than from human annotation or physical ray-tracing, our results show that HAL-Net learns to approximate that heuristic, not that the

heuristic itself matches real photochemical halation. We address this directly in Section 6.

3 Related Work

Learned film artifact analysis. The closest prior work is FGA-NN (Ameur et al., 2025), which trains a neural network to estimate film grain parameters so grain can be stripped before video compression and re-synthesized after decoding. FGA-NN is part of a broader line of work at InterDigital that includes a style-based grain analysis framework and a style-conditioned follow-up, all targeting grain rather than other analog artifacts. FGA-NN and HAL-Net share the same structural idea: regress a small set of synthesis parameters from a CNN rather than generate pixels directly. They differ in what they target. Grain is a global noise process; halation is a local, highlight-anchored glow. No prior learned model addresses halation.

Classical bloom and glow rendering. Outside film emulation, bloom and glow effects have been standard in real-time graphics and photography software for years, typically implemented as a bright-pass threshold followed by a multi-scale Gaussian blur composited back onto the image. Photo editing tools that offer a “halation” or “glow” filter expose this pipeline directly as sliders for blur radius, strength, and tint, with no estimation step: the user supplies parameters from scratch for every image. HAL-Net’s renderer follows the same bright-pass-and-blur structure, but with predicted parameters rather than manual ones.

Parameter regression for photo effects. A body of work trains CNNs to predict a small vector of editing parameters (exposure, white balance, tone curve control points) rather than full pixel output, keeping the effect interpretable and editable after prediction. HAL-Net uses this same pattern: a regression head outputting three bounded scalars through a sigmoid activation, following the approach in FGA-NN and automatic photo-retouching networks.

4 Method

4.1 Overview

HAL-Net has three components, executed in order. A rule-based labeling procedure scans highlight regions of each training image and outputs three pseudo-labels: radius, intensity, and warmth. A compact CNN trains to regress those same three parameters from the raw image, without access to the labeling rules at inference time. A deterministic renderer converts any (radius, intensity, warmth) triple back into a halation overlay on the input. Only the CNN runs at inference time; the labeling procedure produces training targets, and the renderer is a fixed function shared between ground truth and predictions.

4.2 Parameter Definitions

The three halation parameters map to how a user would reason about the effect:

- **Radius** controls how far the glow spreads from a highlight edge. Stored as a normalized value in $[0, 1]$, mapped to a Gaussian blur standard deviation between 0 and 32 pixels at 256x256 resolution.
- **Intensity** controls how strong the glow is relative to the surrounding image, independent of spread distance.
- **Warmth** controls color, from neutral white (0.0) to the saturated red-orange of CineStill 800T (1.0).

All three come from the network’s sigmoid output and are always in $[0, 1]$, so they plug directly into the demo’s slider positions without rescaling.

4.3 Pseudo-Label Extraction

No public dataset pairs photographs with ground-truth halation parameters, so we derive labels from 100 curated CineStill scans via a heuristic that searches for a highlight-anchored warm halo and measures its properties. The procedure runs per image in three stages.

Highlight detection. For each of four decreasing luma thresholds (0.92, 0.80, 0.68, 0.55), we threshold the luma channel to get a candidate highlight mask, then discard the threshold if the mask covers more than 25% of the frame (the scene is broadly bright rather than point-lit) or contains fewer than 4 pixels. A compactness check rejects masks that are large in bounding-box extent but sparsely filled, which screens out diffuse sky glow or wide window blowout rather than discrete practical lights.

Multi-scale radius search. For each surviving mask, we blur it with Gaussian kernels at six candidate radii (sigma 3, 6, 10, 15, 22, and 32 pixels) and inspect the ring-shaped region where the blurred mask is non-trivial but the original mask is not. At each radius we compute a combined warmth-and-bloom signal: 70% weight on how much redder the ring is than the image’s global average (corrected for the same comparison in blue), and 30% weight on how much the blurred mask has spread into that ring. The radius maximizing this combined signal across all four threshold levels becomes the image’s best radius estimate.

Intensity and warmth measurement. Given the selected radius, intensity is computed from the brightness difference between the halo ring and the rest of the image, scaled by the warmth signal at that radius, and clipped to $[0.05, 1.0]$. Warmth is the excess red versus excess blue in the halo ring relative to global channel means, normalized into $[0, 1]$. A separate edge-based detector runs in parallel for halation that hugs a long thin edge (a neon tube, a window rim) rather than a compact point source. This path requires a dark-field scene, a low shadow floor (to screen out misty or foggy shots), and a warm overall color temperature (blue-channel mean below red-channel mean). When both paths fire, their estimates are blended, with the point-source intensity capped relative to the edge path’s. Images where neither path fires receive a floor label (radius 0.05, intensity 0.05, warmth 0.2) so the network trains on clear negative examples too.

4.4 Dataset

The labeling procedure runs over 100 CineStill 800T scans from Flickr and Lomography, hand-selected for visible halation around practical light sources: street lamps, neon signs, candles, and windows. Images are resized to 256x256 and split 70/15/15 into train, validation, and test sets with a fixed random seed. Every label comes from the heuristic above, not a human annotator. The dataset is best understood as a way to distill a hand-designed detector into a fast network, not as independently collected ground truth.

4.5 Network Architecture

The feature extractor has three stride-2 convolutional layers that progressively downsample a 256x256x3 input while widening channels (32, 64, then 128), followed by two residual blocks at 128 channels, each with two 3x3 convolutions, batch normalization, and a skip connection. Adaptive average pooling collapses the feature map to a 128-dimensional vector. A two-layer regression head (128 to 64 with ReLU and dropout $p=0.3$, then 64 to 3) maps this to the three output parameters, with a final sigmoid bounding each to $[0, 1]$. The full network has 693k parameters and trains in under five minutes on a T4 GPU.

4.6 Training Setup

We train with mean squared error between predicted and pseudo-label parameters, using Adam at $lr=1e-3$ and batch size 8 for 30 epochs. A ReduceLROnPlateau scheduler with patience 5

lowers the rate when validation loss stalls. We checkpoint on each new validation minimum and report results from that best checkpoint.

4.7 Halation Renderer

The renderer is a fixed, non-learned function shared between ground-truth visualization and model predictions. Given an image and a (radius, intensity, warmth) triple, it computes a soft highlight mask by linearly ramping luma between 0.65 and 1.0, then blurs this mask with a Gaussian of standard deviation $\max(\text{radius} * 32 * \text{scale}, 1)$ pixels, where $\text{scale} = \max(H, W) / 256$ so the glow spreads proportionally on high-resolution inputs. The normalized bloom map is added back onto the channels with different weights: the red channel receives the full intensity scaled by a warmth-dependent boost, the green channel receives a small fixed fraction, and the blue channel receives a fraction that shrinks to zero as warmth approaches 1. The glow shifts from a neutral bloom at low warmth to the saturated red-orange of CineStill 800T at high warmth.

5 Experiments

5.1 Setup

We train HAL-Net for 30 epochs on the 70-image training split, monitor validation loss on the 15-image validation split, and evaluate on the 15-image test split using the best validation checkpoint. All results compare HAL-Net predictions against pseudo-label parameters from the heuristic in Section 3.3. We call the heuristic output “ground truth” as shorthand throughout, matching its role as the training target, while acknowledging the caveat from Section 3.4.

5.2 Training Dynamics

Training loss drops from roughly 0.10 at initialization to a noisy floor near 0.04-0.05 by epoch 15 and holds there through epoch 30. Validation loss is noisier with only 15 images across 2 batches: it falls from 0.11 to roughly 0.07 in the first few epochs, spikes around epoch 9 (the scheduler responds), then settles between 0.070 and 0.074 for most remaining epochs, reaching its minimum of 0.066 around epoch 27-28. The train-val gap that opens after epoch 15 reflects mild overfitting, which is expected at this dataset size. The validation plateau, rather than continued increase, shows the gap stabilizes.

5.3 Quantitative Results

Table 1. Per-parameter mean absolute error on the 15-image test split.

Method	Radius MAE	Intensity MAE	Warmth MAE	Mean MAE
Mean-pred baseline	0.012	0.317	0.328	0.219
HAL-Net	0.015	0.247	0.218	0.160

The mean-prediction baseline predicts the training-set mean for every test image. It is the minimum sanity check: a model that cannot beat it has learned nothing. HAL-Net beats it by 0.070 on intensity and by 0.110 on warmth. Radius is the exception: both methods sit near zero because the pseudo-label distribution for radius is very narrow (nearly every image maps to a spread around 0.09), so the training mean is already nearly optimal for every test image. This is a labeling artifact, not evidence that radius is unimportant to the visual result.

5.4 Qualitative Results

Figure 2 compares the original photograph, the heuristic ground-truth halation, and HAL-Net’s predicted halation on three test images, chosen to exclude ceiling cases where intensity is pinned near its maximum. Across all three rows, HAL-Net places the glow in the right location at roughly the right spatial extent: the fire escape railing, the traffic lights and sunset sky, and the rim-lit hair and jacket in the portrait all receive a warm halo. Radius estimates are close throughout (0.06-0.10 predicted versus 0.09-0.10 ground truth). Warmth predictions are consistently lower than ground truth (0.82-0.90 versus 0.99-1.00) but still land toward the orange end of the scale. The main visible gap is intensity: HAL-Net predicts 0.51-0.65 against ground-truth values of 0.50-0.75, with the largest shortfall on the two examples with the strongest glow. The portrait, whose ground-truth intensity sits near the dataset’s mid-range, shows the smallest gap.

5.5 Coverage-Cutoff Ablation

The 25% highlight-coverage gate is the most consequential constant in the pseudo-labeling heuristic: when a luma threshold causes more than 25% of the image to light up, that threshold is skipped on the grounds that the scene is broadly bright rather than point-lit. We re-ran the labeling procedure at 0.15 and 0.40 on the 15-image test split to check whether 0.25 is doing meaningful work.

At a cutoff of 0.15, 10 of the 15 test images collapse to the floor label (intensity 0.05) because the stricter gate rejects most threshold levels even for images with visible halation. At the default 0.25, 9 images collapse, but the 6 that do not include the highest-intensity examples in the test set (up to 1.0). Relaxing to 0.40 reduces floor-label images to 7 and adds 3 images in the 0.9-1.0 intensity range without changing the low-to-mid distribution much. The direction is predictable: a looser gate admits brighter scenes, raising median label intensity while adding noise from broadly overexposed shots that do not carry local halation. This puts 0.25 near-optimal for the current dataset and points to more training images at the high end, not a different cutoff, as the path to closing the intensity gap.

5.6 Discussion

Accurate radius and warmth with under-predicted intensity at the high end follows from the training dynamics. With 70 training images, high-intensity examples are scarce. MSE loss does not penalize under-prediction on rare high-intensity cases more than it penalizes the same error elsewhere, so the model has little reason to commit to an aggressive intensity prediction when a conservative, average-case prediction already lowers aggregate training loss. Radius and warmth come from more stable, low-level statistics (ring location and red-versus-blue channel bias) that a CNN with strided convolutions and residual blocks picks up from a small dataset. The ablation in Section 4.5 shows the intensity gap is a data problem, not an architecture problem.

6 Demo and Interactive Use

HAL-Net runs as a Gradio application with a live instance on Hugging Face Spaces. A user uploads any image; the app pads it to square, resizes it to 256x256, runs it through the network, and renders the predicted halation through the function in Section 3.7. The resolution-aware sigma scaling in the renderer means the glow spreads proportionally on high-resolution inputs rather than appearing tight. The interface shows the three predicted parameters alongside the rendered output, and the renderer is the same one used for ground-truth visualization in Section 4.4, so manual override sliders modify the image through exactly the same path the model’s predictions take. HAL-Net works as a fast initial estimate with a manual fallback, which

addresses the core limitation of existing bloom-based tools discussed in Section 2: no estimation step.

7 Limitations

Ground truth is heuristic, not measured. Every label comes from the rule-based detector in Section 3.3, not from calibrated physical measurement or human annotation. HAL-Net is a neural approximation to that heuristic; evaluation against the heuristic’s own output cannot show how closely either matches real photochemical halation.

Small dataset from a single stock. With 70 training images from CineStill 800T only, the model sees a limited range of halation appearances. Section 4.5 traces the intensity under-prediction to the scarcity of high-intensity training examples. Generalization to other warm-toned analog stocks or digitally simulated halation has not been tested.

No perceptual evaluation. The quantitative results in Section 4.3 measure how well the network approximates the heuristic, not how the rendered output looks to a viewer. A more complete evaluation would include a human preference study between predicted and ground-truth renders.

Renderer simplicity. The renderer in Section 3.7 is a single Gaussian bloom with a fixed channel-mixing rule. It does not model the directional streaking, lens-dependent falloff, or multi-source interaction that real film halation can show around complex light arrangements.

8 Conclusion

HAL-Net applies FGA-NN’s parameter-regression strategy to halation, training a CNN to predict radius, intensity, and warmth from an image’s highlight regions. With no labeled halation dataset available, we built one by distilling a hand-designed highlight-and-halo heuristic into pseudo-labels over 100 CineStill 800T scans, then trained HAL-Net to reproduce those labels from raw pixels in a single forward pass. On held-out test images, HAL-Net beats the mean-prediction baseline on intensity (MAE 0.247 vs 0.317) and warmth (MAE 0.218 vs 0.328), getting the glow location, spatial extent, and warm color cast right at the cost of moderate under-prediction on the strongest examples. A coverage-cutoff ablation shows this gap needs more high-intensity training images, not architecture changes. The model ships as an interactive demo with manual override sliders.

9 References

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